

BanglaMix: A Dataset and Benchmark for Code-Switching ASR in Bangladeshi Regional Dialects

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Abstract—Automatic speech recognition (ASR) for low-resource, dialectal, and code-switched speech is still a challenging problem. Existing Bengali datasets usually focus on either regional dialects or Bengali–English code-switching, but not both, even though speakers in Bangladesh often mix English words into regional dialect speech in daily communication. In this work, we introduce BanglaMix, to our knowledge the first large-scale dataset for dialectal Bengali–English code-switched ASR, containing 41,499 transcribed audio clips (about 131 hours) across 15 regional dialects, collected from social media where natural code-switching is common. We build a fully automated data pipeline using voice activity detection, Whisper-based silver-standard transcription, script-based code-switch detection, and a large language model (LLM) filter to remove uncertain cases, avoiding manual transcription. We evaluate three ASR model families—Whisper, Meta MMS, and wav2vec2-XLS-R—under both zero-shot and fine-tuned settings, and also study different training setups including dialect-only, code-switch-only, and combined data. The fine-tuned Whisper-small model achieves a word error rate (WER) of 0.80 and a character error rate (CER) of 0.49, which is a 25% relative improvement over the zero-shot model and is statistically significant ($p < 0.001$, paired bootstrap) compared to all zero-shot baselines and the fine-tuned CTC model, including zero-shot Whisper large-v3. Although performance improves, error rates remain high due to non-standard dialect spelling and silver-standard labels, and manual evaluation of 61 clips shows label quality of WER 0.70 and CER 0.39. Therefore, BanglaMix provides a challenging open benchmark for future research, and the dataset and code are publicly available¹.

Index Terms—Automatic speech recognition, Bengali ASR, code-switching, dialectal speech, low-resource speech recognition, Whisper, and a large language model judge.

I. INTRODUCTION

Speech is the most natural interface to technology, yet automatic speech recognition (ASR) performance remains highly uneven across languages and speaking styles. Two major factors that significantly degrade ASR are dialectal variation and code-switching (CS), the practice of mixing words from a second language within a single utterance. Both phenomena are pervasive in everyday Bangladeshi speech:

the country has many mutually distinct regional dialects, and younger, educated speakers frequently insert English words into dialectal Bengali, producing utterances such as “*chatgaiya bhashay phone unboxing kortachi, honestly speaking camera performance onek bhalo*” (“unboxing a phone in the Chittagong dialect; honestly speaking, the camera performance is very good”).

Despite this reality, existing Bengali speech resources only address one of these phenomena at a time. Read-speech corpora such as OpenSLR-37 [1] focus on standard Bengali, while the Ben-10 corpus [2] covers ten dialects but contains no code-switching. The MUCS shared task [3] provides Bengali–English code-switching data, but only in standard (Dhaka) Bengali. As a result, no dataset or model captures the intersection of regional dialects and English code-switching, even though this is one of the most common forms of informal speech in Bangladesh. This mismatch between real-world speech and available resources creates a fundamental barrier for robust ASR development.

Recognizing dialectal code-switched speech is challenging due to several compounding factors. First, regional dialects differ significantly from standard Bengali in vowel quality, consonant clusters, prosody, and vocabulary, causing acoustic models trained on standard Bengali to generalize poorly. Second, code-switching requires models to seamlessly transition between Bengali and English within a single utterance, often involving mixed scripts that monolingual systems cannot handle. Third, dialectal and code-switched speech is underrepresented in existing corpora, and large-scale manual transcription across 15 dialects is prohibitively expensive for a low-resource language setting. Finally, dialect words and borrowed English terms often lack standardized spelling, meaning that even linguistically correct transcriptions can be penalized under strict word-level evaluation metrics.

These challenges jointly motivate the design of our dataset, pipeline, and evaluation strategy. Specifically, they influence our decision to adopt an automated annotation pipeline without manual transcribers, to use script-aware code-switch detection, to select a shared-vocabulary sequence-to-sequence modeling framework, and to emphasize character-level and code-switch-

¹Dataset: https://huggingface.co/datasets/niloycste68/Bangali_local_dialect_ASR_HF_Dataset, Code: <https://github.com/niloycste/ASR-Bangali-Local-Dialect->

aware evaluation metrics.

This work fills this gap; Fig. 1 provides an end-to-end overview of the dataset construction, models, and evaluation. The main contributions of this paper are as follows:

- 1) To our knowledge, the first large-scale dataset for dialectal Bengali–English code-switched ASR, containing 41,499 transcribed clips (about 131 hours) across 15 regional dialects, collected from social media using dialect-name search queries.
- 2) An end-to-end automated pipeline that does not require human annotators, using voice-activity segmentation, Whisper-based silver-standard transcription, script-based code-switching detection, and a strong multilingual LLM as a judge for borderline cases to enable large-scale annotation.
- 3) A systematic benchmark and ablation study comparing three ASR model families under zero-shot and fine-tuned settings, including controlled training subsets (dialect-only, code-switching-only, and combined) to measure the contribution of each data type.
- 4) A detailed evaluation and error analysis across dialects and code-switching conditions, including statistical significance testing, to identify failure patterns and establish the dataset as a challenging benchmark for current ASR systems.

Beyond dataset construction, this work also highlights an important empirical finding: high error rates are not incidental but reflect the intrinsic difficulty of this speech setting. Even large-scale pretraining (e.g., models trained on hundreds of thousands of hours of audio) struggles to achieve strong performance on dialectal code-switched Bengali. This suggests that such speech remains largely outside the effective coverage of current ASR systems, leaving millions of speakers underserved. Therefore, our primary contribution is not only the dataset itself but also the establishment of a measurable benchmark for this previously unquantified performance gap, supported by a carefully validated silver-label noise estimate (Section III-E). The goal is not to achieve state-of-the-art performance, but to expose a realistic and challenging setting that can guide future research toward more inclusive ASR systems.

The rest of the paper is organized as follows. Section II reviews related work on Bengali, dialectal, and code-switched ASR. Section III describes the BanglaMix dataset and its construction process. Section IV presents the ASR models and fine-tuning strategy, while Section V details the experimental setup. Section VI reports the main results, Section VII provides detailed error analysis, ablation studies, and case studies, Section VIII discusses limitations and reproducibility, and Section IX concludes the paper with future directions.

Table I summarizes the notation used in this paper.

II. RELATED WORK

This section reviews prior work related to Bengali automatic speech recognition, including standard and dialectal ASR systems, code-switching speech recognition, large-scale pre-trained speech models, automated speech corpus construction,

TABLE I
SUMMARY OF NOTATION.

Symbol	Description
η_{lo}, η_{hi}	low/high no-speech acceptance thresholds (0.30, 0.60)
n_{bn}, n_{en}	number of Bengali / English words in a clip
b	number of intra-utterance language boundaries
ρ	English-word ratio $n_{en}/(n_{en} + n_{bn})$
S, D, I	substitutions, deletions, insertions in the alignment
N	number of reference tokens
P, R	precision and recall of code-switch detection
$\mathcal{B}$	CTC alignment-collapsing function

and large language model-based data quality evaluation. Although substantial progress has been made in multilingual and low-resource speech recognition, most existing approaches focus on standard Bengali speech and do not adequately address the combined challenges of regional dialect variation and code-switched speech.

A. Bengali and Dialectal ASR

Early Bengali ASR relied on a Gaussian mixture and a deep neural network hidden Markov models with hand-built pronunciation lexicons [4], [5], which were limited to standard Bengali and struggled with out-of-vocabulary words. The field then moved to end-to-end models: connectionist temporal classification (CTC) [6], deep recurrent networks [7], Deep Speech [8], and attention-based encoder–decoders such as Listen, Attend, and Spell [9], supported by toolkits such as ESPnet [10], which lowered error rates on standard Bengali [11]. The Transformer [12] and masked pre-training [13] subsequently enabled large self-supervised and weakly-supervised speech models—wav2vec 2.0 [14], HuBERT [15], WavLM [16], the multilingual XLS-R [17], [18], Whisper [19], and MMS [20]—which, together with multilingual corpora such as Common Voice [21], improved transfer to low-resource Bengali. Dialectal Bengali received focused attention only recently. The Ben-10 corpus [2] covers ten regional dialects (over 78 hours, 394 speakers) and shows that even strong foundation models, including Whisper large-v3, perform poorly on out-of-distribution dialect speech, both zero-shot and after fine-tuning. Biswas *et al.* [22] propose a WavLM-based denoising and adaptation framework for dialectal Bengali (e.g., Chittagonian and Sylheti) under simulated noise, reporting large relative WER reductions over wav2vec 2.0 and Whisper at low signal-to-noise ratios. Ridoy *et al.* [23] compare Whisper and Wav2Vec-BERT on standard Bangla and find the self-supervised encoder is both more accurate and more computationally efficient under low-resource conditions. None of these, however, addresses code-switching, which motivates a benchmark that combines dialect and code-switching in a single resource.

B. Code-Switching ASR

Code-switching—alternating between two languages within a conversation or even a single utterance—is pervasive in

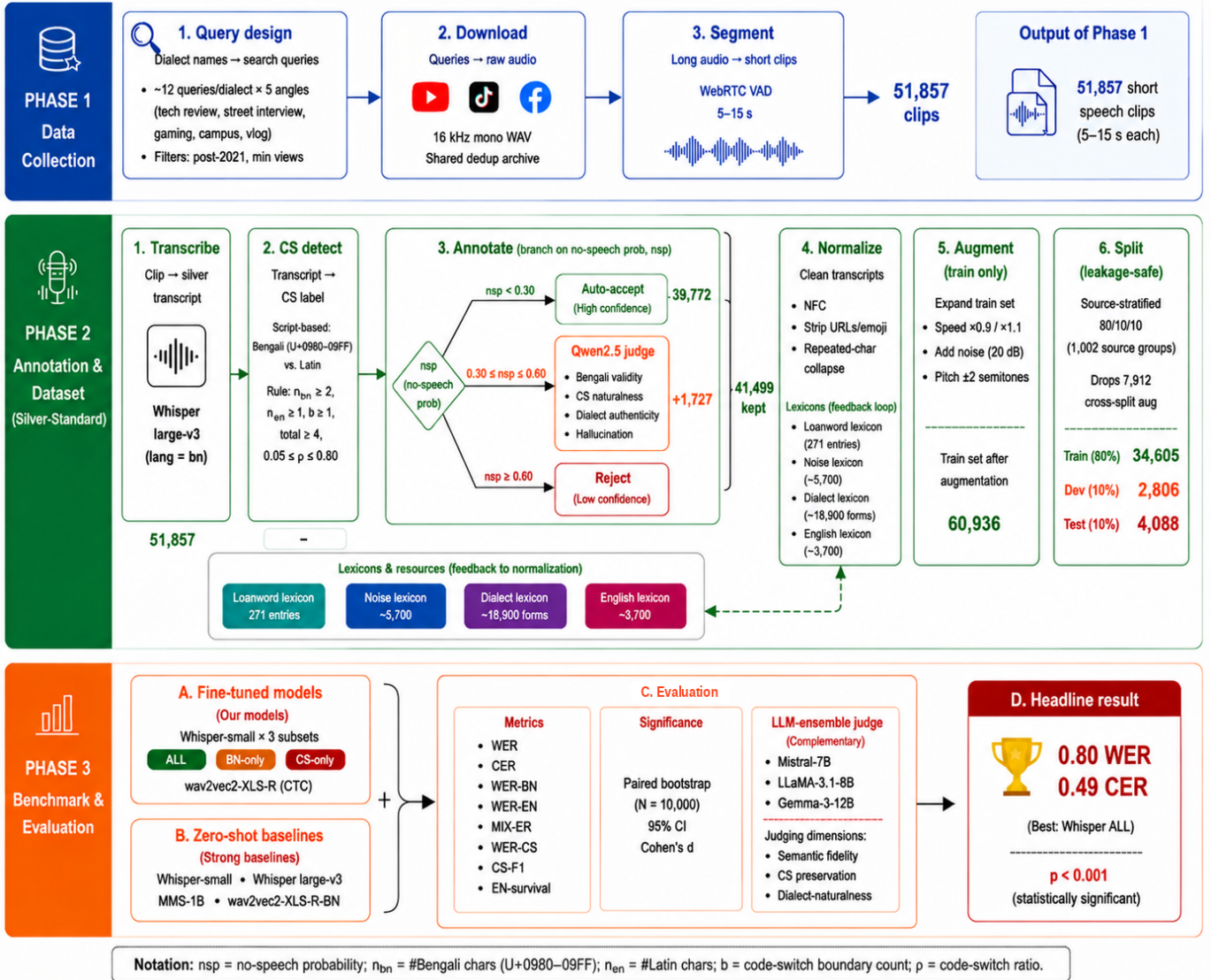


Fig. 1. End to end overview of BanglaMix. Dialect-targeted social media collection and voice activity segmentation feed a fully automated silver-standard labeling pipeline (Whisper large-v3 transcription, script based code-switch detection, and an LLM-based quality judge), producing 41,499 transcribed clips. After normalization and a leakage-safe, source-stratified split, we benchmark fine-tuned and zero-shot ASR models using edit-distance, code-switch-aware, and LLM-based evaluation metrics.

multilingual societies and especially common in informal Bangladeshi speech, where English words are routinely embedded in Bengali sentences. It is hard for ASR for several compounding reasons: the effective vocabulary spans two languages and, for Bengali–English, two scripts; code-switched utterances are rare in standard training corpora; and language switches interact with accent and acoustic variation, complicating both the acoustic and language-modelling components [24], [25]. Dedicated corpora have driven much of the progress in the field, most prominently the Mandarin–English SEAME corpus [26], while recent analysis shows that even large multilingual models handle code-switching unreliably unless given explicit code-switched supervision [27]. For Indian languages, the MUCS 2021 challenge [3] introduced

Bengali–English and Hindi–English code-switching benchmarks, but its Bengali portion is restricted to standard (Dhaka) Bengali and contains no regional dialects. A useful property of the Bengali–English pair is that the two languages occupy disjoint Unicode ranges, so word-level language identification is almost exact from script alone, without a dedicated language-identification model. Architecturally, sequence-to-sequence models such as Whisper are well-suited to code-switching because their shared sub-word vocabulary [28] already covers both scripts and can emit mixed-script text in a single decoding pass, whereas monolingual CTC models such as MMS must commit to one output script and cannot represent code-switching at all. This work evaluates both paradigms in order to quantify that gap on dialectal Bengali.

C. Automated Corpus Construction for Low-Resource ASR

Because manual transcription does not scale to many low-resource varieties, recent corpora are built with automated pipelines. GigaSpeech 2 [29] assembles a large multi-domain corpus for Thai, Indonesian and Vietnamese through automated web crawling, Whisper-based transcription, and iterative refinement, demonstrating that weakly supervised labels can replace human transcribers at scale. Complementary work uses synthetic speech and spectrogram augmentation to enlarge low-resource ASR training pools [30]. BanglaMix follows this automated philosophy but targets a previously uncovered setting—multi-dialect Bengali with English code-switching—and adds an LLM judge for borderline transcripts.

D. LLM-as-a-Judge for Data Quality

Using large language models (LLMs) as automatic evaluators has become common in natural-language processing, ranging from pairwise response judging [31], [32] to multi-dimensional quality scoring [33]. Such judges are attractive precisely when human annotation is costly, and reference labels are scarce or noisy—as in low-resource dialectal ASR—although their reliability depends on prompt design and the capacity of the judging model. In this work, an LLM judge is used at two distinct stages. During *dataset construction*, a single strong multilingual model (Qwen2.5 [34]) scores borderline silver-standard transcripts along several quality dimensions and decides whether to retain them. During *evaluation*, an ensemble of three open models—Mistral [35], LLaMA [36], and Gemma [37]—provides an auxiliary, reference-free signal that complements word-level metrics (Section VII). Applying LLM judges to both quality control and evaluation for code-switched, low-resource ASR has, to the authors’ knowledge, not been explored before.

E. Positioning of This Work

Table II summarizes representative Bengali and code-switching ASR resources along the dimensions most relevant to this work. Two gaps are clear. Standard-Bengali corpora and models [1], [4], [5], [11] address neither dialect nor code-switching. The dialectal corpus Ben-10 [2] covers multiple dialects but no code-switching, while the MUCS task [3] covers code-switching but only in standard Bengali. General multilingual models [17], [19], [20] transfer to standard Bengali but degrade sharply on dialects and, for monolingual CTC models, cannot emit code-switched output at all. BanglaMix is, to our knowledge, the first resource to occupy the intersection: *multi-dialect Bengali with natural English code-switching*.

III. THE BANGLAMIX DATASET CONSTRUCTION

The BanglaMix dataset is created to study real-world Bengali speech that includes different regional dialects and natural English code-switching. The main goal is to collect speech that reflects how people actually speak in everyday situations. The dataset focuses on three key ideas: using real dialect speech, capturing natural code-switching, and building the dataset without manual transcription. All data is collected from social

media using dialect-based search queries to ensure authentic regional speech instead of standard pronunciation. After collection, the audio is processed using automatic segmentation, transcription, and quality checking methods. This results in a large dataset that can be used to test and improve speech recognition systems for challenging real-world conditions.

A. Design Principles and Why They Matter

BanglaMix is built around three principles. First, *dialectal authenticity*: the audio must contain genuine regional dialect, not standard Bengali spoken by someone from a region. This matters because region-name queries (for example, “Chittagong vlog”) mostly retrieve people *from* a region speaking standard Bengali to a general audience; dialect-name queries instead surface content where creators deliberately use the dialect. Second, *natural code-switching*: CS must arise spontaneously in informal speech rather than from scripted content, because scripted CS does not reflect real acoustic and lexical patterns. Third, *scalability*: the pipeline must be fully automated so the dataset can grow without paid transcribers, which is essential for a low-resource language where annotation budgets are small.

B. Query Design

Search queries are the single most important lever for collecting authentic dialectal code-switched speech, because they determine *which* creators are retrieved. As noted above, region-name queries surface standard-Bengali content, so this work instead uses *dialect-name* queries combined with content genres in which spontaneous code-switching is most frequent. For each dialect, around a dozen queries are designed across five content angles, summarized in Table III. The rationale is that street interviews and live-reaction content are spontaneous and cannot sustain a fake accent, technology and campus content elicit English technical terms, and vlog content yields relaxed, mixed, informal speech. Quality filters exclude pre-2021 uploads and videos with negligible viewership.

C. Construction Pipeline

Algorithm 1 summarizes the end-to-end pipeline; each step is justified below.

Step 1: acquisition and segmentation: Audio is downloaded at 16 kHz mono using `yt-dlp` [38] from social-media platforms (YouTube, TikTok, and Facebook), with a shared de-duplication archive so the same video is never collected twice. Long recordings are segmented into 5–15 s clips using WebRTC voice-activity detection [39]. Segmentation is necessary because raw social-media audio contains long non-speech regions, and fixed 5–15 s clips match the input length that modern ASR models train on, which keeps both transcription and fine-tuning efficient. This step yields 51,857 clips.

Step 2: transcription and CS detection: Each clip is transcribed with Whisper large-v3 [19] with the language forced to Bengali. Whisper large-v3 is chosen as the labeler because it is the strongest openly available multilingual model and therefore the best available proxy for a human transcriber on this hard

TABLE II

COMPARATIVE SUMMARY OF REPRESENTATIVE BENGALI AND CODE-SWITCHING ASR RESOURCES AND MODELS. “CS” INDICATES CODE-SWITCHING SUPPORT; “–” DENOTES NOT APPLICABLE. BANGLAMIX IS THE FIRST TO COMBINE MULTI-DIALECT BENGALI WITH ENGLISH CODE-SWITCHING.

Work	Method / Model	Objective	Variety	CS	Hours	Key limitation
Hasnat et al. [4]	GMM-HMM	Isolated-word Bengali ASR	Standard BN	No	–	Lexicon-dependent; WER >50%; no OOV handling
Swarna et al. [5]	Survey (HMM/GMM/DNN)	Bengali phoneme recognition survey	Standard BN	No	–	Survey; standard Bengali, no dialect or CS
Mandal et al. [11]	End-to-end (CTC/attention)	Lexicon-free Bengali ASR	Standard BN	No	–	Standard Bengali only; no dialect or CS
Kjartansson et al. [1]	OpenSLR-37 corpus	Read-speech benchmark	Standard BN	No	100	Read speech; no dialect, no code-switching
MUCS 2021 [3]	CS challenge corpus	Bengali-English CS-ASR	Standard BN	Yes	16	Standard Dhaka Bengali; no dialect
Ben-10 [2]	Whisper fine-tuning	Multi-dialect Bengali ASR	Dialectal BN	No	78	No code-switching
XLS-R [17]	wav2vec2 self-supervised	Multilingual transfer	Multilingual	No	–	Standard Bengali via transfer; no dialect/CS
Whisper [19]	Weakly-supervised seq2seq	Multilingual ASR	Multilingual	No	–	~70% WER on unseen Bengali dialects
MMS [20]	CTC with adapters	1000+ language ASR	Multilingual	No	–	Monolingual output; cannot emit code-switching
Ridoy et al. [23]	Whisper vs. Wav2Vec-BERT	Low-resource Bangla model comparison	Standard BN	No	~90–100	Standard Bangla only; no dialect or CS
Biswas et al. [22]	WavLM denoising + adaptation	Noise-robust dialectal Bengali ASR	Dialectal BN	No	~20	Dialectal but no code-switching; noise-focused
GigaSpeech 2 [29]	Automated crawl + transcribe + refine	Large-scale low-resource corpus	Th/Id/Vi	No	~30k	Not Bengali; no dialect or code-switching

TABLE III

CONTENT ANGLES USED TO DESIGN DIALECT-NAME SEARCH QUERIES, AND WHY EACH ELICITS AUTHENTIC DIALECTAL CODE-SWITCHING.

Content angle	Why it elicits dialectal CS
Technology/phone review	Forces English technical terms within dialect speech
Street interview/reaction	Spontaneous public responses; a fake accent cannot be sustained
Gaming / live stream	Live commentary with frequent English game terms
Campus/career	Academic and job English mixed into regional speech
Vlog / content creator	Relaxed informal speech with frequent CS

data. Code-switching is detected directly from Unicode script ranges: Bengali (U+0980–U+09FF) and Latin (ASCII) occupy disjoint ranges, so word-level language identification is nearly exact without a dedicated model. Let n_{bn} and n_{en} be the number of Bengali- and Latin-script words in a clip, b the number of language boundaries (adjacent words of different scripts), and $\rho = n_{en}/(n_{en} + n_{bn})$ the English-word ratio. A clip is labeled code-switched when

$$(n_{bn} + n_{en}) \geq 4 \wedge n_{bn} \geq 2 \wedge n_{en} \geq 1 \wedge b \geq 1 \wedge 0.05 \leq \rho \leq 0.80. \quad (1)$$

A stricter high-confidence subset additionally requires $n_{en} \geq 2$ and $0.10 \leq \rho \leq 0.70$. The lower bound on ρ excludes iso-

Algorithm 1 BanglaMix: From raw dialectal audio to a curated CS-ASR benchmark

Input: dialect-name queries $\mathcal{Q}$; VAD aggressiveness a ; no-speech thresholds $\tau_{lo}=0.30$, $\tau_{hi}=0.60$; split ratio 80/10/10
Output: curated dataset $\mathcal{D} = (\mathcal{D}_{tr}, \mathcal{D}_{dev}, \mathcal{D}_{te})$

Step 1: Acquire and segment

- 1: $\mathcal{R} \leftarrow \text{DOWNLOAD16K}(\mathcal{Q})$ ▷ yt-dlp + ffmpeg; shared dedup archive
- 2: $\mathcal{C} \leftarrow \text{VADSEGMENT}(\mathcal{R}, a)$ ▷ WebRTC VAD; 5–15 s clips → 51,857

Step 2: Transcribe and detect code-switching

- 3: **for all** clip $c \in \mathcal{C}$ **do**
- 4: $t_c \leftarrow \text{WHISPERLARGEV3}(c, \text{lang}=\text{bn})$
- 5: $c.\text{CS} \leftarrow \text{SCRIPTCSDetect}(t_c)$ ▷ Bengali vs. Latin Unicode ranges
- 6: **end for**

Step 3: Silver-standard annotation (no human transcribers)

- 7: $\mathcal{A} \leftarrow \{c : \text{NSP}(c) < \tau_{lo}\}$ ▷ auto-accept → 39,772
- 8: $\mathcal{M} \leftarrow \{c : \tau_{lo} \leq \text{NSP}(c) < \tau_{hi}\}$ ▷ borderline (middle band) → 8,712
- 9: **for all** $c \in \mathcal{M}$ **do**
- 10: $d \leftarrow \text{QWEN2.5JUDGE}(t_c)$ ▷ 4 dimensions; accept/reject/uncertain
- 11: **if** $d = \text{ACCEPT}$ **then**
- 12: $\mathcal{A} \leftarrow \mathcal{A} \cup \{c\}$ ▷ +1,727
- 13: **end if**
- 14: **end for**

Step 4: Normalize transcripts

- 15: **for all** $c \in \mathcal{A}$ **do**
- 16: $t_c \leftarrow \text{NORMALIZE}(t_c)$ ▷ NFC; variant lexicon; repetition collapse
- 17: **end for**

Step 5: Augment (training pool only)

- 18: $\mathcal{G} \leftarrow \text{AUGMENT}(\mathcal{A})$ ▷ speed, noise, pitch

Step 6: Quality filter and leakage-safe split

- 19: $\mathcal{A} \leftarrow \text{DROPLowQUALITY}(\mathcal{A} \cup \mathcal{G})$
- 20: $(\mathcal{D}_{tr}, \mathcal{D}_{dev}, \mathcal{D}_{te}) \leftarrow \text{SOURCESTRATSPPLIT}(\mathcal{A}, 80/10/10)$ ▷ per-source; dialect-stratified
- 21: $\mathcal{D}_{tr} \leftarrow \mathcal{D}_{tr} \cup \text{PINAUGToTRAIN}(\mathcal{G})$
- 22: **return** $\mathcal{D}$ ▷ final curated dataset

lated borrowed words and the upper bound excludes English-dominant clips, so that only genuinely mixed utterances are tagged; the same rule is applied to the model hypotheses when

computing code-switch detection F1.

Step 3–4: silver-standard annotation with an LLM judge: Manual transcription of dialectal CS speech at scale is infeasible, so a two-stage silver-standard strategy is used. High-confidence transcripts (Whisper no-speech probability below 0.30) are accepted directly, following evidence that high-confidence Whisper output is a usable training signal [29], [40]; clips above 0.60 are rejected as noise. The uncertain middle band is the hard case, and here a single strong multilingual judge (Qwen2.5 [34]) scores each transcript on Bengali validity, code-switch naturalness, dialect authenticity, and absence of hallucination, and returns accept/reject/uncertain. A single high-capacity judge is used at this stage—rather than an ensemble—because the borderline set is large and a single strong model offers the best quality-per-compute trade-off; running it locally also avoids API cost and keeps the data private. After this stage, 41,499 clips remain.

Step 5: normalization: Transcripts are cleaned with Unicode NFC normalization, removal of zero-width characters, URLs, hashtags, mentions, and emoji, digit normalization, repeated character collapse, and punctuation/whitespace cleanup. A second, evaluation-only step maps spelling variants of borrowed English words to a single canonical form using a curated lexicon of 272 entries (650 variant spellings). This is important because dialectal Bengali has *no standardized orthography* for loanwords, so the same spoken word may be written in several valid ways. Without this mapping, the word error rate would be inflated by legitimate spelling variation rather than genuine recognition errors. The pipeline additionally maintains three supporting lexicons that encode dialect- and code-switch-specific knowledge: a list of about 5,700 known ASR hallucination and noise tokens, used to detect and remove corrupted transcripts; a list of about 18,900 valid dialectal Bengali word forms that are explicitly *protected* from being normalized toward standard spellings, so genuine dialect vocabulary is preserved; and a list of about 3,700 common English words that supports the script-based identification of code-switched tokens. These lexicons make the otherwise generic normalization sensitive to the two phenomena—regional dialect and code-switching—that this dataset targets.

Step 6–7: augmentation, quality filter, and leakage-safe split: Audio augmentation (speed perturbation, additive noise, pitch shift) is applied to the training pool to improve robustness to acoustic variation, complementing spectrogram-level methods such as SpecAugment [41]. The data is then split at the *source* level—all clips from one original video stay in one split—to prevent speaker and channel leakage, which would otherwise inflate test scores. The split is dialect-stratified, so every dialect appears in all three partitions. Augmented clips inherit their source and are pinned to training only when that source is in the training split, so augmentation never leaks a source across splits.

D. Statistics

Table IV lists the 15 dialects, and Fig. 2 shows the clip distribution, which is intentionally broad but naturally imbal-

TABLE IV
DIALECT COVERAGE IN THE RELEASED DATASET.

Dialect	Clips	Hours
Sylhet	7,931	26.0
Old Dhaka	4,741	14.6
Sandwip	4,583	13.0
Barishal	3,259	11.0
Chittagong	2,993	9.1
Khulna	2,774	8.9
Narail	2,371	7.8
Kishoreganj	2,287	7.1
Rangpur	1,985	6.4
Comilla	1,847	6.1
Narsingdi	1,661	5.1
Habiganj	1,446	4.5
Mymensingh	1,369	4.1
Noakhali	1,140	3.6
Tangail	1,112	3.8
Total	41,499	131.2

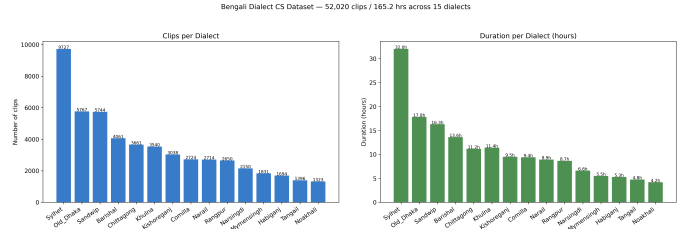


Fig. 2. Clip distribution across the 15 dialects.

anced, mirroring online content availability. Of the 51,857 segmented clips (≈ 164 hours of raw audio), 41,499 (131.2 hours) pass silver-standard annotation and constitute the released dataset—the remaining ≈ 33 hours are clips rejected by the no-speech threshold or the LLM judge; Table V reports its train/dev/test splits. The dataset’s primary scale is its dialect breadth—15 varieties over 131 hours—while code-switching forms a deliberate *natural-prevalence* minority (5.7% of clips): English insertion is a spontaneous, in-the-wild phenomenon, and we capture it at its real frequency rather than artificially oversampling it. Code-switching is therefore treated as a distinct evaluation axis layered on top of broad dialect coverage, and CS-specific results (Section VI) are reported on this naturally small subset accordingly.

E. Human Validation of Silver-Standard Labels

Because the transcripts are silver-standard, we quantify their reliability against human reference. A bilingual annotator transcribed 61 randomly drawn test clips (2–6 per dialect, balanced across the code-switching axis) by listening to the audio; the human transcript is treated as the reference, and the silver label as the hypothesis. Overall agreement is WER 0.70 and CER 0.39. The large WER–CER gap is informative: at the character level, the labels are far closer to human transcription than the word-level figure suggests, reflecting the absence of

TABLE V
TRAIN/DEV/TEST SPLITS OF THE RELEASED (REAL, UNAUGMENTED) DATASET. AUDIO AUGMENTATION EXPANDS THE TRAINING SPLIT TO 60,936 CLIPS FOR THE MODEL TRAINING, DEV, AND TEST REMAIN REAL AND UNAUGMENTED.

Split	Clips	Hours	CS clips	CS%
Train	34,605	108.9	2,019	5.8
Dev	2,806	9.1	166	5.9
Test	4,088	13.2	176	4.3
Total	41,499	131.2	2,361	5.7

TABLE VI
HUMAN VALIDATION OF SILVER-STANDARD LABELS ON 61 TEST CLIPS (HUMAN REFERENCE VS. SILVER HYPOTHESIS). LOWER IS BETTER; n IS THE NUMBER OF CLIPS PER DIALECT.

Dialect	n	WER	CER
Sandwip	3	0.26	0.10
Kishoreganj	5	0.46	0.25
Khulna	3	0.53	0.32
Old Dhaka	6	0.54	0.31
Sylhet	4	0.54	0.34
Narail	3	0.63	0.27
Barishal	5	0.72	0.39
Tangail	4	0.74	0.41
Rangpur	6	0.75	0.51
Narsingdi	3	0.76	0.38
Chittagong	4	0.80	0.40
Mymensingh	6	0.84	0.47
Comilla	2	0.92	0.44
Noakhali	3	1.04	0.59
Habiganj	4	1.06	0.60
Overall	61	0.70	0.39

a standardized orthography for these dialects—annotators and Whisper frequently choose different spellings of the same spoken word, which inflates WER without indicating a semantic error.

Two patterns are consistent across the sample. First, code-switched labels are *more* reliable than Bengali-only ones (WER 0.58 vs. 0.74): the embedded English tokens act as clear acoustic anchors that the transcription model captures well. Second, label quality varies by dialect (Table VI): Sandwip, Kishoreganj, Sylhet, Khulna, and Old Dhaka are transcribed reliably (WER ≤ 0.54 , CER ≤ 0.34), whereas Habiganj and Noakhali are the weakest (WER > 1.0 , CER ≈ 0.6), driven by Whisper insertions and hallucination on their more divergent phonology. These figures bound the silver-label noise floor, contextualize the absolute error rates reported in Section VI, and motivate label-refinement directions (e.g., judge-guided self-training) for future work. Per-dialect estimates rest on small samples ($n = 2\text{--}6$) and are therefore indicative rather than definitive.

F. Ethical and Legal Considerations

All audio is sourced from publicly available social-media content. To respect creators and platform terms, the released dataset will distribute segmented audio clips and transcripts

for non-commercial research use, together with the source identifiers and timestamps so that the original content can be located and attribution preserved. Because the data is conversational and may contain personal opinions, no speaker identities, demographic labels, or contact information is collected or released, and the dataset is intended only for building and evaluating speech-recognition systems. Dialect coverage follows online content availability rather than population, so it is intentionally not a demographic sample; this imbalance is reported transparently (Fig. 2) so that downstream users can weight or resample as needed.

Several further considerations apply. *Consent and platform terms*: the source content is public, but its creators did not explicitly consent to research use; we therefore release the data strictly for non-commercial research, honor platform Terms of Service and copyright by distributing short derived segments with source identifiers rather than re-hosting whole videos, and we provide a contact-based *takedown/opt-out* process so any creator can request removal of their content. *Voice as personal data*: speech is biometric and potentially identifying, so we release no speaker names, handles, faces, or contact information, segment the audio into short clips that hinder the reconstruction of full recordings, and explicitly prohibit speaker-identification, surveillance, or re-identification uses in the accompanying license. *Potential harms*: as With any ASR resource, downstream models could be misused or could underperform for less-represented dialects; we mitigate this through per-dialect reporting, transparent disclosure of label noise (Section III-E), and an intended-use statement limiting the dataset to building and evaluating ASR for these underserved varieties. A formal ethics board review was not required for the use of public, de-identified media under the host institution’s policy, but the above safeguards were applied throughout.

IV. ASR MODELS AND FINE-TUNING

This work evaluates three model families, chosen to cover the two dominant ASR paradigms and a range of pre-training regimes.

Whisper (sequence-to-sequence): Whisper [19] is an encoder-decoder transformer trained on 680k hours of weakly supervised multilingual audio. It is central to this work because its sub-word vocabulary contains both Bengali and Latin tokens so that it can emit mixed-script (code-switched) output in a single decoding pass—a natural fit for the task. While convolution-augmented architectures such as the Conformer [42] are also competitive for ASR, Whisper’s shared multilingual vocabulary makes it especially suited to code-switching. The fine-tuned models use Whisper-small (244M parameters), which is feasible to train on modest hardware while still benefiting from large-scale pre-training. All inference forces Bengali transcription.

Meta MMS (CTC, multilingual): MMS [20] is a CTC model covering 1,000+ languages with per-language adapters. It is included as a strong multilingual zero-shot baseline; because its output vocabulary is monolingual Bengali, it

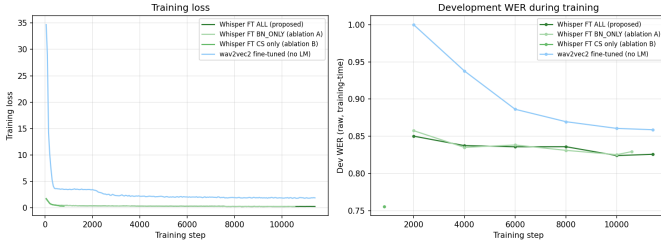


Fig. 3. Training loss (left) and development WER (right) during fine-tuning. The development WER is the raw training-time value and is higher than the final normalized test WER in Section VI.

cannot emit English script, which makes it a useful contrast for measuring how much code-switching specifically hurts.

wav2vec2-XLS-R (CTC, self-supervised): wav2vec2-XLS-R-300M [17] is pre-trained on 436k hours of multilingual audio. A Bengali-fine-tuned checkpoint is used as a zero-shot baseline, and it is additionally fine-tuned on BanglaMix to represent the CTC paradigm under the same training data as Whisper.

Fine-tuning procedure: Whisper is fine-tuned with the Hugging Face Transformers [43] and Datasets [44] libraries using the Adam optimizer [45]: learning rate 10^{-5} , effective batch size 16, three epochs, 500 warm-up steps, with the best checkpoint selected by development WER. For the CTC model, a character vocabulary covering Bengali, Latin, and common punctuation is built from the training transcripts, the CTC head is reinitialized, and the encoder is fine-tuned at learning rate 10^{-4} for three epochs. All fine-tuning and benchmark inference (zero-shot baselines and fine-tuned systems) run on a single NVIDIA A100-SXM4-40GB GPU, whereas the one-time silver-standard transcription of all clips with Whisper large-v3 was performed on a CPU workstation (Intel Core i9-14900, 24 cores, 32 GB RAM). The CTC model is trained to minimize the negative log-likelihood marginalized over all alignments π that collapse to the target y ,

$$\mathcal{L}_{\text{CTC}} = -\log \sum_{\pi \in \mathcal{B}^{-1}(y)} \prod_t p(\pi_t | x) \quad (2)$$

where $\mathcal{B}$ is the CTC collapsing function, whereas Whisper is trained with the standard token-level cross-entropy of its decoder. Key hyperparameters are listed in Table VII. Fig. 3 shows that all Whisper variants converge within about 2k steps; the CS-only ablation runs for fewer steps because it uses only the roughly 4.5k code-switched clips.

Training subset ablation: To isolate the contribution of each data type, three Whisper variants are trained: BN_ONLY (dialect-only clips), CS_ONLY (code-switched clips only), and ALL (both). All three share the same base model, procedure, and test set, so any difference is attributable to the training data composition.

V. EXPERIMENTAL SETUP

This section explains how the experiments are conducted to evaluate the proposed models. It describes the evaluation

TABLE VII
FINE-TUNING HYPERPARAMETERS FOR THE TWO TRAINED MODEL FAMILIES.

Hyperparameter	Whisper-small	wav2vec2-XLS-R
Base parameters	244M	300M
Architecture	seq2seq	CTC
Learning rate	10^{-5}	10^{-4}
Effective batch size	16	16
Epochs	3	3
Warm-up steps	500	500
Precision	bf16/fp16	bf16/fp16
Model selection	best dev WER	best dev WER

metrics, baseline systems, and statistical testing methods. The main goal is to fairly measure model performance on dialectal and code-switched speech. Different metrics are used to capture different types of errors, including word-level errors, character-level errors, and code-switching errors. Standard baseline models are used for comparison, and statistical tests are applied to ensure that the results are reliable.

A. Metrics and Why Each is Used

We use Word Error Rate (WER) and Character Error Rate (CER) as the main evaluation metrics. CER is more important in this work because dialectal Bengali does not have a fixed spelling system. Because of this, WER can wrongly penalize correct answers that are just spelled differently, while CER better reflects how accurately the speech is recognized.

WER is further divided into WER-BN and WER-EN, which measure errors separately for Bengali and English words. This helps us understand where the model performs well or poorly. We also use Mixed Error Rate (MIX-ER), which checks whether both word meaning and script (Bengali or English) are correctly recognized together. In addition, Code-Switch Detection F1 (CS-F1) is used to measure whether the model can correctly detect if code-switching is present in the output.

Before evaluation, all outputs are normalized using the same rules described in Section III, including removing repeated or noisy text patterns. Standard evaluation tools are used for all calculations [46], [47].

Formally, WER is calculated using substitutions, deletions, and insertions:

$$\text{WER} = \frac{S + D + I}{N} \quad (3)$$

where S , D , and I represent substitution, deletion, and insertion errors, and N is the total number of words.

CER is calculated in the same way but at the character level:

$$\text{CER} = \frac{S_c + D_c + I_c}{N_c} \quad (4)$$

Here, CER is more reliable for dialectal speech because spelling can vary a lot even when the pronunciation is correct.

WER-BN and WER-EN measure errors separately for Bengali-script words and English-script words. MIX-ER

TABLE VIII

MAIN RESULTS ON THE BANGLAMIX TEST SET (LOWER IS BETTER EXCEPT CS-F1). **BOLD GREEN** MARKS THE BEST OVERALL SYSTEM; “–” DENOTES A METRIC NOT APPLICABLE TO A MONOLINGUAL CTC MODEL.

Model	WER	CER	WER-BN	WER-EN	MIX-ER	WER-CS	CS-F1
<i>Zero-shot baselines</i>							
Whisper-small	1.07	1.02	1.07	1.26	1.15	1.02	0.00
Whisper large-v3	0.91	0.76	0.92	1.08	0.98	0.80	0.14
MMS-1B	0.99	0.57	0.99	1.11	–	0.94	0.04
wav2vec2-XLS-R-BN	0.96	0.62	0.96	1.03	–	0.94	0.04
<i>CTC fine-tuned on BanglaMix</i>							
wav2vec2	0.86	0.49	0.86	1.00	0.97	0.88	0.14
<i>Whisper fine-tuned on BanglaMix (ablation)</i>							
BN_ONLY (A)	0.80	0.49	0.80	1.00	0.94	0.82	0.01
CS_ONLY (B)	0.88	0.55	0.89	0.76	0.82	0.81	0.28
ALL (proposed)	0.80	0.49	0.80	0.93	0.91	0.82	0.19

checks whether both the word and its language (Bengali or English) are correctly predicted.

For code-switch detection, we treat it as a yes/no problem: whether code-switching appears in the output or not. We use Precision, Recall, and F1 score:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad CS-F1 = \frac{2PR}{P + R} \quad (5)$$

Finally, English-word survival (EN-Surv) measures how many English words in the reference are correctly preserved in the model output for code-switched speech:

$$EN-Surv = \frac{\sum_{(r,h) \in \mathcal{P}} |\{w \in r : \text{English}(w)\} \cap h|}{\sum_{(r,h) \in \mathcal{P}} |\{w \in r : \text{English}(w)\}|} \quad (6)$$

where $\mathcal{P}$ is the set of code-switched (reference, hypothesis) pairs.

B. Baselines

We compare our method with four zero-shot baseline models. These include Whisper-small (used as a base model), Whisper large-v3 (the strongest open ASR model), MMS-1B, and wav2vec2-XLS-R-BN, which is trained on standard Bengali speech.

C. Significance Testing

To ensure results are reliable, we use a paired bootstrap test with 10,000 resamples over the test set. For each sample, we compute WER and then estimate confidence intervals. This method is appropriate because all models are tested on the same dataset, making the comparison fair and consistent.

VI. RESULTS

Table VIII summarizes the main experimental results, while Fig. 4 presents an overall comparison of word error rate (WER) across all evaluated models. The results indicate clear performance differences between zero-shot models, task-adapted systems, and the proposed approach under dialectal and code-switched speech conditions.

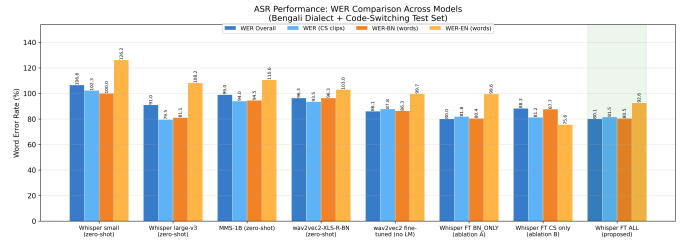


Fig. 4. Word error rate across all models. Fine-tuned systems are clearly separated from zero-shot baselines.

Performance of zero-shot models: Zero-shot models show limited robustness under dialectal and code-switched conditions. In particular, Whisper large-v3 achieves a WER of 0.91 despite being trained on large-scale multilingual data (680k hours). Whisper-small yields a higher WER of 1.07. These observations suggest that large-scale multilingual pre-training alone may not be sufficient for reliable generalization to dialectal and code-switched Bengali speech.

Effect of task-specific fine-tuning: Fine-tuning on BanglaMix leads to noticeable improvements in performance. Whisper-small reduces WER from 1.07 to 0.80 and CER from 1.02 to 0.49, corresponding to an approximate 25% relative reduction in WER. This indicates that dataset-specific adaptation may play an important role beyond model scale for this task.

Overall system performance: The proposed ALL configuration achieves the most balanced performance across evaluation metrics, with a WER of 0.80 and CER of 0.49. It outperforms all zero-shot baselines and the fine-tuned wav2vec2 model (0.86 WER), with statistical significance at $p < 0.001$ (paired bootstrap; Section VII). Compared with Whisper large-v3, consistent improvements are observed across all reported error metrics. However, the difference between ALL and the BN-only configuration is not statistically significant in terms of overall WER ($p = 0.54$), suggesting that gains may primarily arise from improved handling of code-switching rather than

TABLE IX
PER-DIALECT WER/CER FOR THE PROPOSED WHISPER ALL MODEL.

Dialect	Clips	WER	CER
Sylhet	1,111	0.93	0.65
Old Dhaka	360	0.78	0.53
Sandwip	176	0.92	0.67
Barishal	282	0.87	0.54
Chittagong	198	0.79	0.46
Khulna	120	0.66	0.39
Kishoreganj	743	0.75	0.36
Comilla	157	0.78	0.42
Narail	464	0.70	0.41
Rangpur	165	0.79	0.51
Narsingdi	60	0.72	0.39
Mymensingh	56	0.70	0.35
Habiganj	75	0.90	0.52
Tangail	51	0.74	0.39
Noakhali	70	0.70	0.36
Overall	4,088	0.80	0.49

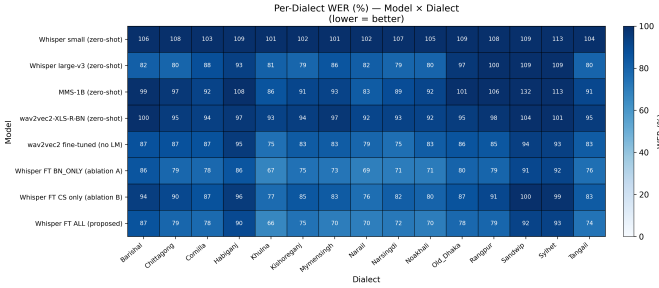


Fig. 5. Model $\times$ dialect WER heatmap. Darker is higher WER.

general recognition accuracy.

Ablation analysis: The ablation results indicate trade-offs between overall recognition accuracy and code-switching capability. The BN-only model achieves competitive WER (0.80) but exhibits limited ability to handle code-switching, as reflected by a CS-F1 of 0.01. In contrast, the CS-only model improves code-switching related metrics (CS-F1 = 0.28; WER-EN = 0.76), but shows degraded overall WER (0.88). The proposed ALL model provides a more balanced performance across both general recognition and code-switching behavior.

A. Per-Dialect Results

Table IX reports per-dialect performance for the proposed model, while Fig. 5 shows the corresponding WER heatmap across dialects and systems. The results indicate noticeable variability across dialects, with relatively higher error rates for more phonetically divergent varieties.

Table X compares representative models across dialects. The proposed system achieves the best or tied-best performance in 14 out of 15 dialects. A slight advantage of Whisper large-v3 is observed in one dialect (Barishal), which may indicate that model scale still plays a role in certain acoustic conditions.

TABLE X
PER-DIALECT WER FOR REPRESENTATIVE MODELS. **BOLD GREEN** = BEST PER DIALECT.

Dialect	large-v3	wav2vec2	Whisper ALL
Sylhet	1.09	0.93	0.93
Old Dhaka	0.97	0.86	0.78
Sandwip	1.09	0.94	0.92
Barishal	0.82	0.87	0.87
Chittagong	0.80	0.87	0.79
Khulna	0.81	0.75	0.66
Kishoreganj	0.79	0.83	0.75
Comilla	0.88	0.87	0.78
Narail	0.82	0.79	0.70
Rangpur	1.00	0.85	0.79
Narsingdi	0.79	0.75	0.72
Mymensingh	0.86	0.83	0.70
Habiganj	0.93	0.95	0.90
Tangail	0.80	0.83	0.74
Noakhali	0.80	0.83	0.70

TABLE XI
ERROR-TYPE RATES AND ENGLISH-WORD SURVIVAL (EN-SURV).

Model	S	D	I	EN-Surv
Whisper large-v3 (zs)	0.31	0.48	0.12	0.65
Whisper-small (zs)	0.25	0.75	0.07	0.12
MMS-1B	0.66	0.13	0.19	0.00
wav2vec2 (zs)	0.52	0.36	0.08	0.00
wav2vec2 (FT)	0.52	0.23	0.10	0.00
Whisper BN_ONLY	0.45	0.21	0.14	0.01
Whisper CS_ONLY	0.53	0.19	0.16	0.37
Whisper ALL	0.45	0.21	0.14	0.13

VII. ANALYSIS AND DISCUSSION

This section provides a detailed analysis of model behavior beyond aggregate error metrics. It examines error decomposition, code-switching performance, and qualitative failure cases across models and dialects. In addition, it evaluates code-switch structure preservation, compares performance against human references, and analyzes statistical significance. The goal is to better understand where improvements arise and which aspects of dialectal and code-switched ASR remain challenging.

A. Error Types and English-Word Survival

Table XI breaks WER into substitution (S), deletion (D), and insertion (I) rates, and reports English-word survival (EN-Surv), the fraction of English reference words that appear in the hypothesis on CS clips. Fig. 6 visualizes the S/D/I breakdown.

A clear pattern emerges. Monolingual CTC models (MMS, wav2vec2) and the Bengali-only ablation essentially never reproduce English words (EN-Surv at most 0.01); they delete English or map it to the Bengali script. Code-switch training raises this sharply—CS_ONLY reaches 0.37, the best among fine-tuned models, and ALL retains 0.13 while keeping the lowest overall WER. Zero-shot Whisper large-v3 preserves English best of all (0.65), but at a far higher overall WER (0.91), as it captures neither dialect phonology nor a coherent transcript. This confirms that sequence-to-sequence models

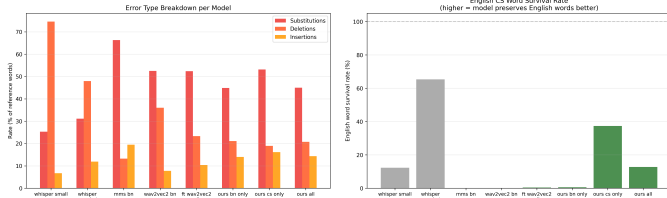


Fig. 6. Substitution/deletion/insertion rates per model.

TABLE XII
WER ON CS VS. BENGALI-ONLY CLIPS. GAP = WER-CS − WER-BN.

Model	WER-CS	WER-BN	Gap
Whisper large-v3 (zs)	0.80	0.92	−0.12
Whisper-small (zs)	1.02	1.07	−0.05
MMS-1B	0.94	0.99	−0.05
wav2vec2 (FT)	0.88	0.86	+0.02
Whisper CS_ONLY	0.81	0.89	−0.08
Whisper ALL	0.82	0.80	+0.02

with a shared multilingual vocabulary, fine-tuned on code-switched data, handle CS words far more faithfully than monolingual CTC models.

B. Where Code-Switching Difficulty Lives

We first note the scope of the CS evidence: the per-clip CS metrics (WER-CS, CS-F1) are computed on the 176 code-switched test clips and are therefore *indicative* rather than high-precision. The central CS finding, however, does not depend on that sample size: it is the *qualitative* English-deletion effect—monolingual and Bengali-only models drop English words almost entirely—which is a large, unambiguous effect visible across thousands of English tokens (Table XI), not a marginal difference on a small subset. Consistent with this, a paired bootstrap ($N=10,000$) on the 176 CS clips yields a wide 95% confidence interval for the proposed model’s WER-CS, $[0.76, 0.87]$, that overlaps those of the other Whisper systems (BN_ONLY $[0.78, 0.86]$; zero-shot large-v3 $[0.75, 0.84]$): whole-clip WER-CS does *not* statistically separate models at this sample size, which is precisely why we treat English-word survival—not WER-CS—as the discriminative code-switching measure.

Table XII reports WER on CS versus Bengali-only clips. The whole clip gap is small (within ± 0.12): because CS clips are predominantly Bengali, code-switching does not greatly change overall clip WER. The difficulty instead lives in the English tokens themselves, as Table XI shows—English survival is near zero for monolingual and BN-only models. Fig. 7 shows that models never trained on CS collapse to near-zero CS-F1, and that training on code-switched data is what recovers code-switch behavior.

Beyond *whether* English words survive, we also measure *whether* models reproduce the code-switching *structure*—the number of intra-utterance language switches—computed over

TABLE XIII
CODE-SWITCH STRUCTURE PRESERVATION ON CS CLIPS (WHISPER SEQ2SEQ FAMILY): AVERAGE INTRA-UTTERANCE LANGUAGE SWITCHES PRODUCED, AND THE FRACTION OF REFERENCE SWITCHES PRESERVED. REFERENCE CLIPS AVERAGE 3.48 SWITCHES.

Model	Switches/clip	Switch preserved
Reference (human silver)	3.48	—
Whisper CS_ONLY	2.19	47%
Whisper large-v3 (zs)	1.71	44%
Whisper ALL (proposed)	1.32	33%
Whisper BN_ONLY	0.85	20%

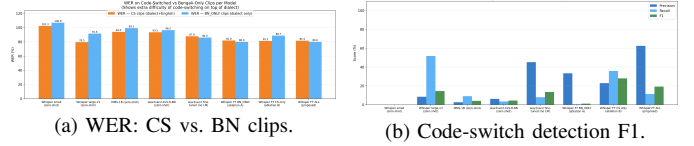


Fig. 7. Code-switching analysis. Models never trained on CS show near-zero CS-F1.

roughly 600 reference switch points (more stable than per-clip WER). We report this for the Whisper seq2seq family, for which code-switching is genuinely representable; monolingual CTC models (MMS) are excluded because they cannot reliably emit Latin script (EN-survival 0.00; Table XI), so any apparent “switches” in their output are character-level artifacts rather than real code-switching. To avoid that artifact, a Latin token is counted as English only when it has at least two letters. Reference CS clips contain, on average, 3.5 switches, yet every Whisper system produces far fewer (Table XIII): models systematically *collapse* code-switching rather than reproduce it. Among the fine-tuned systems, the ordering tracks code-switch supervision—CS-only preserves the most switch structure (47%), Bengali-only the least (20%), and the proposed model lies in between (33%)—confirming from a third independent angle that code-switch training restores switching behavior. Zero-shot large-v3 preserves a comparable fraction (44%), but only by over-generating text: it also shows the highest English survival (Table XI) yet the worst overall WER (0.91), i.e. it keeps switches at the cost of accuracy. The measure counts switch *density* rather than exact positions, so it upper-bounds positional accuracy.

C. Qualitative Error Analysis

Inspecting model outputs reveals three recurring error patterns, illustrated using romanized Bengali examples for readability (Bengali script omitted).

- Repetition hallucination refers to a failure mode where, on out-of-distribution clips, zero-shot Whisper produces degenerate repetitions. For example, a short reference such as “tai” may be expanded into many repeated tokens. This behavior contributes substantially to insertion errors in zero-shot systems and is largely removed by repetition-collapse normalization during scoring.

TABLE XIV

ERROR TAXONOMY: DELETION/INSERTION/SUBSTITUTION SHARE OF ALL WORD ERRORS, AND THE FRACTION OF ALL ERRORS THAT ARE BENIGN SPELLING VARIANTS (NEAR-IDENTICAL SUBSTITUTIONS, CHAR-DISTANCE ≤ 0.34).

Model	Del	Ins	Sub	Spell. var.
Whisper large-v3 (zs)	53%	13%	34%	5%
wav2vec2 (FT)	27%	12%	61%	15%
Whisper ALL	26%	18%	56%	17%

- English deletion refers to a pattern where monolingual and Bengali-focused models systematically drop English words from code-switched utterances. For example, the reference “*ami office-e meeting korlam*” may be decoded as “*ami . . . korlam*”, omitting the English portion. This explains the near-zero English-word survival reported in Table XI.
- Spelling variance refers to cases where fine-tuned models produce phonetically correct outputs with different orthographic forms, such as “*kortasi*” versus “*kortechi*” for the same spoken word. These cases are counted as word-level errors but do not necessarily indicate recognition failure, which is reflected in the substantially lower CER compared to WER.

These patterns explain the main trends in the results: fine-tuning primarily reduces hallucination and English-word deletion, while the remaining errors are largely driven by benign spelling variation due to unstandardized dialectal orthography.

To make this quantitative, Table XIV decomposes every word error into deletions, insertions, and substitutions, and further estimates the share of *all* errors that are benign spelling variants—substitutions whose reference and hypothesis words differ by a normalized character distance ≤ 0.34 (i.e. a near-identical spelling). The contrast is stark: zero-shot large-v3 is dominated by *deletions* (53%)—it simply fails to transcribe large spans of dialect speech—whereas fine-tuning halves the deletion share (to $\sim 26\%$) and converts the output into mostly complete transcripts whose errors are predominantly substitutions, roughly one-sixth of which (15–17% of all errors) are mere spelling variants rather than genuine recognition mistakes. This quantifies why CER is substantially lower than WER for the fine-tuned systems and why WER overstates their true error.

D. Ablation Study

Two ablations isolate key design choices. The *training-subset* ablation (Table VIII) shows that BN_ONLY and CS_ONLY each sacrifice one capability—BN_ONLY collapses on English (CS-F1 0.01) while CS_ONLY loses overall accuracy (0.88 WER)—so only training on all data is strong on both. The *normalization* ablation (Table XV) isolates the repetition-collapse step: applying it lowers the proposed model’s WER from 0.96 to 0.80 and CER from 0.60 to 0.49 *without changing the model*, by preventing Whisper’s degenerate decoding loops from dominating the score.

TABLE XV

EFFECT OF THE REPETITION-COLLAPSE NORMALIZATION ON THE PROPOSED MODEL (MODEL UNCHANGED; SCORING ONLY).

Configuration	WER	CER
Without repetition collapse	0.96	0.60
With repetition collapse (proposed)	0.80	0.49

E. Case Study

A single concrete case illustrates the dominant failure mode. On a short Sylheti test clip whose reference is one word, zero-shot Whisper-small emits the same token roughly forty times in a degenerate loop, producing an enormous per-clip insertion count; the proposed fine-tuned model, decoded with repetition collapse, instead returns a short hypothesis of the correct length. This case explains both why zero-shot insertion rates are high (Table XI) and why the normalization ablation (Table XV) yields such a large apparent gain: a small number of pathological clips, rather than a uniform degradation, drive much of the zero-shot error. The complementary failure on code-switched clips is the deletion of English tokens by monolingual and Bengali-only models, quantified by the near-zero English-word survival in Table XI.

F. Statistical Significance

All WER improvements of the proposed system over the four zero-shot baselines and the fine-tuned CTC model are significant at $p < 0.001$ (paired bootstrap, $N=10,000$). The 95% confidence interval for the proposed system’s WER is $[0.79, 0.81]$, which does not overlap with that of any baseline (for example, zero-shot large-v3, $[0.90, 0.92]$). The proposed model is *not* significantly different from the BN-only ablation on overall WER (0.8009 vs. 0.8004, $p=0.54$); the two are equivalent on this metric, and the benefit of training on all data instead appears in code-switch handling (CS-F1 0.19 vs. 0.01). Effect sizes against the baselines are statistically robust but numerically small (Cohen’s $d = 0.09$ versus large-v3), as expected when comparing models that all operate in a high-error regime on this challenging benchmark.

G. LLM-Ensemble Evaluation

Word error rate captures surface accuracy but not whether a hypothesis is *semantically* faithful or preserves code-switching. As a complementary, reference-free signal, an ensemble of three open LLMs (Mistral-7B [35], LLaMA-3.1-8B [36], Gemma-3-12B [37]) scores each hypothesis from 0–5 on semantic correctness, code-switch preservation, and dialect naturalness, on a stratified 50-clip sample per model. Table XVI reports the means. The proposed Whisper ALL model attains the highest overall score (2.56/5), the best semantic correctness (2.55), and the best code-switch preservation (2.71); zero-shot large-v3 is marginally higher only on dialect naturalness (2.57 vs. 2.43), and the CS-only ablation is lowest overall (2.21). Inter-annotator (inter-judge) agreement among the three LLM judges—the fraction of judge pairs

TABLE XVI

LLM-ENSEMBLE EVALUATION (MEAN SCORE 0–5; THREE OPEN JUDGES; STRATIFIED 50-CLIP SAMPLE PER MODEL). **BOLD GREEN** = BEST PER COLUMN.

Model	Sem.	CS-pres.	Dialect	Overall
Whisper large-v3	2.41	2.46	2.57	2.48
Whisper CS_ONLY	2.27	2.32	2.05	2.21
Whisper ALL	2.55	2.71	2.43	2.56

whose per-clip scores agree within one point on the 0–5 scale—is 0.73–0.85, indicating the ensemble is internally consistent rather than arbitrary. This corroborates the WER findings under an independent, semantics-aware metric—the proposed model is judged best, and its advantage is largest precisely on code-switch preservation—while the mid-range absolute scores reflect the same task difficulty discussed above.

H. Why Absolute Error Rates are High

A WER near 0.80 is high relative to clean ASR (typically 0.05–0.20). Three factors explain this and frame how the numbers should be read. First, the task is intrinsically hard: 15 low-resource dialects combined with code-switching. Second, dialectal Bengali has no standardized spelling, so word-level WER penalizes valid spelling variants—which is why CER (0.49) is the more informative metric here. Third, the reference labels are silver-standard (generated by Whisper large-v3), so part of the measured error reflects label noise rather than model failure. For these reasons, BanglaMix is positioned as a *challenging open benchmark* rather than a solved task, and the relative, statistically significant gains are the central result.

I. Evaluation Against Human References

Because the test labels are silver-standard, the main table scores each model against Whisper-generated references, which can favor Whisper-family systems. As a bias-free check, we re-score every model against the *human* transcripts on the 61 human-validated clips (Section III-E); Table XVII reports the result. The silver-label ranking partially reverses: against human references, the zero-shot MMS-1B model attains the lowest WER (0.76) and CER (0.41), whereas the proposed Whisper ALL system—best under silver-label scoring on this subset (0.69 WER; cf. 0.80 on the full test set)—rises to 0.83. This is the expected signature of label-source bias: silver labels generated by Whisper inflate the apparent quality of Whisper-family hypotheses, and removing that bias with human references exposes it. Two observations keep the picture coherent. First, fine-tuning still helps *within* the Whisper family even on human references (Whisper ALL 0.83 vs. zero-shot Whisper-small 1.03 and large-v3 0.90), so the value of training on BanglaMix is real. Second, MMS-1B’s lead is consistent with its much larger (1B) multilingual backbone versus the 244M Whisper-small used here, echoing the per-dialect evidence that backbone scale matters (Section VI). We therefore report results under both reference types and present BanglaMix as an open benchmark rather than crowning a single system. These figures rest on a small sample ($n=61$, 14 code-switched)

TABLE XVII

MODEL ERROR AGAINST HUMAN REFERENCES VS. SILVER LABELS ON THE 61 HUMAN-VALIDATED TEST CLIPS. WER(HUMAN) IS THE BIAS-FREE, GOLD-REFERENCE NUMBER; WER(SILVER) IS ON THE SAME CLIPS FOR COMPARISON. LOWER IS BETTER.

Model	WER _{human}	CER _{human}	WER _{silver}
MMS-1B (zero-shot)	0.76	0.41	0.89
Whisper ALL (proposed)	0.83	0.49	0.69
Whisper BN_ONLY	0.84	0.52	0.70
wav2vec2 (fine-tuned)	0.85	0.48	0.78
wav2vec2-XLS-R-BN (zero-shot)	0.87	0.55	0.92
Whisper CS_ONLY	0.88	0.52	0.81
Whisper large-v3 (zero-shot)	0.90	0.72	0.81
Whisper-small (zero-shot)	1.03	0.99	1.04

and are indicative; the *direction* follows directly from the bias mechanism and is unlikely to be an artifact.

VIII. LIMITATIONS

Several limitations frame the scope of this work and should guide the interpretation of the results. First, the transcripts are *silver-standard*: they are produced by Whisper large-v3 and validated by an LLM judge rather than by human annotators, so part of the measured error may reflect label noise. Human validation on 61 clips (Section III-E) estimates this noise at WER 0.70 / CER 0.39 and shows that it varies across dialects (WER 0.26–1.06). Because the silver labels originate from Whisper large-v3, the benchmark may also slightly favor Whisper-family systems over other architectures; therefore, a larger human-validated subset would be useful for more reliable per-dialect estimates. Second, the fine-tuned models use Whisper-small for tractability, and larger backbones would likely reduce absolute error, as suggested by the per-dialect comparison for dialects such as Barishal. Third, code-switched clips are a minority of the dataset (~6%), reflecting their natural prevalence, although more targeted query design could increase their proportion. Fourth, dialect coverage follows online content availability and is therefore imbalanced, with Sylhet dominating; as broader low-resource ASR studies note, such imbalance may under-serve speakers of less-represented language varieties, making per-dialect reporting and fairness-aware evaluation important [48]. Finally, the data is drawn from informal social-media domains such as vlogs, comedy, and reviews, so models trained on BanglaMix may not directly transfer to formal domains such as broadcast news, healthcare, or education.

IX. CONCLUSION

This work introduces BanglaMix, to the best of our knowledge, the first large-scale dataset and benchmark for dialectal Bengali–English code-switching ASR, together with a fully automated, annotator-free construction pipeline that enables systematic evaluation of ASR systems under realistic multi-dialect and code-switched speech conditions. The experimental results across three model families indicate that zero-shot multilingual models perform poorly on this task, while fine-tuning on BanglaMix yields substantial and statistically

significant improvements. The analysis further suggests that code-switching errors are primarily concentrated in English tokens rather than in overall utterance-level recognition errors. The proposed fine-tuned Whisper-based system achieves 0.80 WER and 0.49 CER, outperforming all zero-shot baselines as well as the fine-tuned CTC model under the evaluated setting. Despite these improvements, absolute error rates remain relatively high, which is consistent with challenges arising from dialectal variation, non-standard orthography, and silver-standard annotations, and therefore BanglaMix is released as an open benchmark to support further research in this direction. Future work may explore larger backbone models, the creation of a human-validated evaluation subset, and more targeted data collection for underrepresented dialects such as Sylhet, along with n -gram language model shallow fusion for CTC-based systems using KenLM [49] and beam-search decoding [50], as well as further integration of language models and dialect-aware tokenization strategies following recent low-resource language model fusion approaches [51], [52].

DATA AND CODE AVAILABILITY

The BanglaMix dataset, including segmented audio clips, silver-standard transcripts, and train/dev/test split manifests, is publicly available on the Hugging Face Hub at https://huggingface.co/datasets/niloycste68/Bangali_local_dialect_ASR_HF_Dataset. The complete data construction and evaluation codebase is available on GitHub at <https://github.com/niloycste/ASR-Bengali-Local-Dialect->. Both resources are released for non-commercial research use to support reproducibility. The fine-tuned model checkpoints will be released upon acceptance. All experiments are fully reproducible using a single configuration file that fixes model identifiers, hyperparameters, and random seeds, ensuring deterministic data splits and evaluation. Additionally, evaluation applies consistent normalization across all models, and bootstrap significance testing uses a fixed seed, enabling exact replication of reported confidence intervals and p -values from the prediction files.

REFERENCES

- [1] O. Kjartansson, S. Sarin, K. Pipatsrisawat, M. Jansche, and L. Ha, "Crowd-sourced speech corpora for Javanese, Sundanese, Sinhala, Nepali, and Bangladeshi Bengali," in *Proc. 6th Workshop on Spoken Language Technologies for Under-Resourced Languages (SLTU)*, 2018, pp. 52–55. [Online]. Available: <https://www.openslr.org/37/>
- [2] T. Tashwar Dipto, A. Hossain, R. Sabbir Faruque, M. Rezuwan Hassan, K. Fatema, T. Shome, R. Naswan, M. Foriduzzaman Zihad, M. U. Anam, N. Tasnim *et al.*, "Are ASR foundation models generalized enough to capture features of regional dialects for low-resource languages?" *arXiv e-prints*, 2025, arXiv:2510.
- [3] A. Diwan, R. Vaideeswaran, S. Shah, A. Singh, S. Raghavan, S. Khare, V. Unni, S. Vyas, A. Rajpuria, C. Yarra *et al.*, "MUCS 2021: Multilingual and code-switching ASR challenges for low resource Indian languages," in *Proc. Interspeech 2021*, 2021.
- [4] M. A. Hasnat, J. Mowla, and M. Khan, "Isolated and continuous Bangla speech recognition: Implementation, performance and application perspective," in *Proceedings of the International Symposium on Natural Language Processing (SNLP)*, Hanoi, Vietnam, 2007.
- [5] S. T. Swarna, S. Ehsan, M. S. Islam, and M. E. Jannat, "A comprehensive survey on Bengali phoneme recognition," *arXiv preprint arXiv:1701.08156*, 2017.
- [6] A. Graves, S. Fernández, F. Gomez, and J. Schmidhuber, "Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks," in *Proceedings of the 23rd International Conference on Machine Learning (ICML)*. ACM, 2006, pp. 369–376.
- [7] A. Graves, A.-r. Mohamed, and G. Hinton, "Speech recognition with deep recurrent neural networks," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2013, pp. 6645–6649.
- [8] A. Hannun, C. Case, J. Casper, B. Catanzaro, G. Diamos, E. Elsen, R. Prenger, S. Satheesh, S. Sengupta, A. Coates, and A. Y. Ng, "Deep speech: Scaling up end-to-end speech recognition," *arXiv preprint arXiv:1412.5567*, 2014.
- [9] W. Chan, N. Jaitly, Q. V. Le, and O. Vinyals, "Listen, attend and spell: A neural network for large vocabulary conversational speech recognition," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2016, pp. 4960–4964.
- [10] S. Watanabe, T. Hori, S. Karita, T. Hayashi, J. Nishitoba, Y. Unno, N. E. Y. Soplin, J. Heymann, M. Wiesner, and N. Chen, "ESPnet: End-to-end speech processing toolkit," in *Interspeech*, 2018, pp. 2207–2211.
- [11] S. Mandal, S. Yadav, and A. K. Rai, "End-to-end Bengali speech recognition," *arXiv preprint arXiv:2009.09615*, 2020.
- [12] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 5998–6008.
- [13] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of NAACL-HLT*, 2019, pp. 4171–4186.
- [14] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [15] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed, "HuBERT: Self-supervised speech representation learning by masked prediction of hidden units," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 3451–3460, 2021.
- [16] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao, J. Wu, L. Zhou, S. Ren, Y. Qian, Y. Qian, J. Wu, M. Zeng, and F. Wei, "WavLM: Large-scale self-supervised pre-training for full stack speech processing," *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1505–1518, 2022.
- [17] A. Babu, C. Wang, A. Tjandra, K. Lakhotia, Q. Xu, N. Goyal, K. Singh, P. von Platen, Y. Saraf, J. Pino, A. Baevski, A. Conneau, and M. Auli, "XLS-R: Self-supervised cross-lingual speech representation learning at scale," in *Proc. Interspeech 2022*, 2022, pp. 2278–2282.
- [18] A. Conneau, A. Baevski, R. Collobert, A. Mohamed, and M. Auli, "Un-supervised cross-lingual representation learning for speech recognition," in *Interspeech*, 2021, pp. 2426–2430.
- [19] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*. PMLR, 2023, pp. 28492–28518.
- [20] V. Pratap, A. Tjandra, B. Shi, P. Tomasello, A. Babu, S. Xu, A. Baevski, A. Conneau, R. Collobert, Y. Saraf, and M. Auli, "Scaling speech technology to 1,000+ languages," in *Proceedings of the 41st International Conference on Machine Learning (ICML)*. PMLR, 2024.
- [21] R. Ardila, M. Branson, K. Davis, M. Henretty, M. Kohler, J. Meyer, R. Morais, L. Saunders, F. M. Tyers, and G. Weber, "Common voice: A massively-multilingual speech corpus," in *Proceedings of the 12th Language Resources and Evaluation Conference (LREC)*, 2020, pp. 4218–4222.
- [22] S. Biswas, T. Sheikh *et al.*, "A unified denoising and adaptation framework for self-supervised Bengali dialectal ASR," *arXiv preprint arXiv:2509.00988*, 2025.
- [23] M. S. I. Ridoy, S. Akter, and M. A. Rahman, "Adaptability of ASR models on low-resource language: A comparative study of Whisper and Wav2Vec-BERT on Bangla," in *2025 2nd International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*. IEEE, 2025, pp. 1–6.
- [24] E. Yilmaz, H. van den Heuvel, and D. A. van Leeuwen, "Acoustic and textual data augmentation for improved ASR of code-switching speech," in *Proc. Interspeech 2018*, 2018, pp. 1933–1937.

- [25] S. Sitaram, K. R. Chandu, S. K. Rallabandi, and A. W. Black, "A survey of code-switched speech and language processing," *arXiv preprint arXiv:1904.00784*, 2019.
- [26] D.-C. Lyu, T.-P. Tan, E. S. Chng, and H. Li, "SEAME: a Mandarin-English code-switching speech corpus in south-east asia," in *Interspeech*, 2010, pp. 1986–1989.
- [27] G. I. Winata, S. Cahyawijaya, Z. Liu, Z. Lin, A. Madotto, and P. Fung, "Are multilingual models effective in code-switching?" in *Proceedings of the Fifth Workshop on Computational Approaches to Linguistic Code-Switching*, 2021, pp. 142–153.
- [28] R. Sennrich, B. Haddow, and A. Birch, "Neural machine translation of rare words with subword units," in *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2016, pp. 1715–1725.
- [29] Y. Yang, Z. Song, J. Zhuo, M. Cui, J. Li, B. Yang, Y. Du, Z. Ma, X. Liu, Z. Wang *et al.*, "GigaSpeech 2: An evolving, large-scale and multi-domain ASR corpus for low-resource languages with automated crawling, transcription and refinement," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025, pp. 2673–2686.
- [30] S. Dhahbi, N. Saleem, S. Bourouis, M. Berrima, and E. Verdú, "End-to-end neural automatic speech recognition system for low resource languages," *Egyptian Informatics Journal*, vol. 29, p. 100615, 2025.
- [31] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. King, H. Zhang, J. E. Gonzalez, and I. Stoica, "Judging LLM-as-a-judge with MT-bench and chatbot arena," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.
- [32] Y. Wang, Z. Yu, Z. Zeng, L. Yang, C. Wang, H. Chen, C. Jiang, R. Xie, J. Wang, X. Xie, W. Ye, S. Zhang, and Y. Zhang, "PandaLM: An automatic evaluation benchmark for LLM instruction tuning optimization," *arXiv preprint arXiv:2306.05087*, 2023.
- [33] J. Fu, S.-K. Ng, Z. Jiang, and P. Liu, "GPTScore: Evaluate as you desire," *arXiv preprint arXiv:2302.04166*, 2023.
- [34] Qwen Team, "Qwen2.5 technical report," *arXiv preprint arXiv:2412.15115*, 2024.
- [35] A. Q. Jiang, A. Sablayrolles, A. Mensch, C. Bamford, D. S. Chaplot, D. de las Casas, F. Bressand, G. Lengyel, G. Lample, L. Saulnier, L. R. Lavaud, M.-A. Lachaux, P. Stock, T. Le Scao, T. Lavril, T. Wang, T. Lacroix, and W. El Sayed, "Mistral 7B," *arXiv preprint arXiv:2310.06825*, 2023.
- [36] A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Yang, A. Fan, A. Goyal, A. Hartshorn, A. Yang, A. Mitra, A. Sravankumar *et al.*, "The Llama 3 herd of models," *arXiv preprint arXiv:2407.21783*, 2024.
- [37] Gemma Team, M. Riviere, S. Pathak, P. G. Sessa, C. Hardin, S. Bhupatiraju, L. Hussenot, T. Mesnard, B. Shahriari, A. Ramé, J. Ferret, P. Liu, P. Tafti, A. Friesen, M. Casbon, S. Ramos, R. Kumar *et al.*, "Gemma 2: Improving open language models at a practical size," *arXiv preprint arXiv:2408.00118*, 2024.
- [38] yt-dlp contributors, "yt-dlp: A feature-rich command-line audio/video downloader," 2021. [Online]. Available: <https://github.com/yt-dlp/yt-dlp>
- [39] Google, "WebRTC voice activity detector," 2012. [Online]. Available: <https://webrtc.org>
- [40] S. Gandhi, "Fine-tuning whisper for low-resource speech recognition," HUGGINGFACE Blog, 2023. [Online]. Available: <https://huggingface.co/blog/fine-tune-whisper>
- [41] D. S. Park, W. Chan, Y. Zhang, C.-C. Chiu, B. Zoph, E. D. Cubuk, and Q. V. Le, "SpecAugment: A simple data augmentation method for automatic speech recognition," in *Interspeech*, 2019, pp. 2613–2617.
- [42] A. Gulati, J. Qin, C.-C. Chiu, N. Parmar, Y. Zhang, J. Yu, W. Han, S. Wang, Z. Zhang, Y. Wu, and R. Pang, "Conformer: Convolution-augmented transformer for speech recognition," in *Interspeech*, 2020, pp. 5036–5040.
- [43] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, J. Davison, S. Shleifer, P. von Platen, C. Ma, Y. Jernite, J. Plu, C. Xu, T. Le Scao, S. Gugger, M. Drame, Q. Lhoest, and A. M. Rush, "Transformers: State-of-the-art natural language processing," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP)*. Association for Computational Linguistics, 2020, pp. 38–45.
- [44] Q. Lhoest, A. V. del Moral, Y. Jernite, A. Thakur, P. von Platen, S. Patil, J. Chaumond, M. Drame, J. Plu, L. Patry, A. McMillan-Major, P. Schmid, S. Gugger, C. Delangue, T. Matussiere, L. Debut, S. Bekman, P. Cistac, T. Goehringer, V. Mustar, F. Lagunas, A. Rush, and T. Wolf, "Datasets: A community library for natural language processing," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP)*. Association for Computational Linguistics, 2021, pp. 175–184.
- [45] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *International Conference on Learning Representations (ICLR)*, 2015.
- [46] A. C. Morris, V. Maier, and P. Green, "jiwer: Evaluate your speech-to-text system with WER, MER, WIL, WIP and CER," 2004. [Online]. Available: <https://github.com/jitsi/jiwer>
- [47] HuggingFace, "Evaluate: A HuggingFace library for reproducible evaluation," 2022. [Online]. Available: <https://github.com/huggingface/evaluate>
- [48] S. Hafiz Imam, B. Sani, D. Ketema Gete, B. Yimam Ahamed, I. Said Ahmad, I. Abdulmumin, S. Muhie Yimam, M. Yahuza Bello, and S. H. Muhammad, "Automatic speech recognition for African low-resource languages: Challenges and future directions," *arXiv e-prints*, 2025, arXiv:2505.
- [49] K. Heafield, "KenLM: Faster and smaller language model queries," in *Proceedings of the Sixth Workshop on Statistical Machine Translation (WMT)*. Association for Computational Linguistics, 2011, pp. 187–197.
- [50] Kensho Technologies, "pyctcdecode: A fast and flexible CTC beam search decoder with KenLM language model support," 2021. [Online]. Available: <https://github.com/kensho-technologies/pyctcdecode>
- [51] X. de Zuazo, E. Navas, I. Saratkaga, and I. Hernández Rioja, "Whisper-LM: Improving ASR models with language models for low-resource languages," *arXiv preprint arXiv:2503.23542*, 2025.
- [52] Z. Li and J. Niehues, "Enhance contextual learning in ASR for endangered low-resource languages," in *Proceedings of the 1st Workshop on Language Models for Underserved Communities (LM4UC)*, 2025, pp. 1–7.